Project: Classification Techniques for Predictive Modelling

📌 Problem Statement

Business Context

INN Hotels Group is experiencing a high rate of booking cancellations and no-shows, which is negatively impacting its revenue and operational efficiency. With the rise of online booking platforms, customers now have greater flexibility to cancel reservations at little or no cost, leading to increased last-minute cancellations.

These cancellations result in multiple challenges for the hotel, including:

- Loss of resources (revenue) when the hotel cannot resell the room.
- Additional costs of distribution channels by increasing commissions or paying for publicity to help sell these rooms.
- Lowering prices last minute, so the hotel can resell a room, resulting in a reduction of the profit margin.
- Human resources to make arrangements for the guests.

Therefore, there is a need to identify the key factors influencing booking cancellations and develop a system to proactively predict such cancellations

🎯 Objective

The primary objective of this project is to build a Machine Learning model that can predict whether a hotel booking is likely to be cancelled or not.

The specific goals include:

- Analysing historical booking data to identify patterns and key factors influencing cancellations
- Performing exploratory data analysis (EDA) to understand customer behaviour and booking trends
- Building and evaluating predictive models to accurately classify bookings as cancelled or not
- Providing actionable insights and recommendations to help the hotel reduce cancellations and optimize revenue management strategies

Data Description:

The data contains the different attributes of customers' booking details. The detailed data dictionary is given below.

Data Dictionary:

- Booking_ID: the unique identifier of each booking
- no_of_adults: Number of adults
- no_of_children: Number of Children
- no_of_weekend_nights: Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel
- no_of_week_nights: Number of weeknights (Monday to Friday) the guest stayed or booked to stay at the hotel
- type_of_meal_plan: Type of meal plan booked by the customer:
    - Not Selected – No meal plan selected
    - Meal Plan 1 – Breakfast
    - Meal Plan 2 – Half board (breakfast and one other meal)
    - Meal Plan 3 – Full board (breakfast, lunch, and dinner)
- required_car_parking_space: Does the customer require a car parking space? (0 - No, 1- Yes)
- room_type_reserved: Type of room reserved by the customer. The values are ciphered (encoded) by INN Hotels Group
- lead_time: Number of days between the date of booking and the arrival date
- arrival_year: Year of arrival date
- arrival_month: Month of arrival date
- arrival_date: Date of the month
- market_segment_type: Market segment designation.
- repeated_guest: Is the customer a repeated guest? (0 - No, 1- Yes)
- no_of_previous_cancellations: Number of previous bookings that were canceled by the customer prior to the current booking
- no_of_previous_bookings_not_canceled: Number of previous bookings not canceled by the customer prior to the current booking
- avg_price_per_room: Average price per day of the reservation; prices of the rooms are dynamic. (in euros)
- no_of_special_requests: Total number of special requests made by the customer (e.g. high floor, view from the room, etc)
- booking_status: Flag indicating if the booking was canceled or not.

Overview of the dataset

The first few rows of the dataset.

```
data.head()
```

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space	room_type_reserved	lead_time	arrival_year	arrival_month	arrival_date	market_segment_type	repeated_guest
0	INN00001	2	0	1	2	Meal Plan 1	0	Room_Type 1	224	2017	10	2	Offline	0
1	INN00002	2	0	2	3	Not Selected	0	Room_Type 1	5	2018	11	6	Online	0
2	INN00003	1	0	2	1	Meal Plan 1	0	Room_Type 1	1	2018	2	28	Online	0
3	INN00004	2	0	0	2	Meal Plan 1	0	Room_Type 1	211	2018	5	20	Online	0
4	INN00005	2	0	1	1	Not Selected	0	Room_Type 1	48	2018	4	11	Online	0

no_of_previous_cancellations	no_of_previous_bookings_not_canceled	avg_price_per_room	no_of_special_requests	booking_status
0	0	65.00000	0	Not_Canceled
0	0	106.68000	1	Not_Canceled
0	0	60.00000	0	Canceled
0	0	100.00000	0	Canceled
0	0	94.50000	0	Canceled

```
data.shape
```

```
(36275, 19)
```

- There are 36275 rows and 19 columns in the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36275 entries, 0 to 36274
Data columns (total 19 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Booking_ID                            36275 non-null  object
 1   no_of_adults                          36275 non-null  int64
 2   no_of_children                        36275 non-null  int64
 3   no_of_weekend_nights                  36275 non-null  int64
 4   no_of_week_nights                     36275 non-null  int64
 5   type_of_meal_plan                     36275 non-null  object
 6   required_car_parking_space            36275 non-null  int64
 7   room_type_reserved                    36275 non-null  object
 8   lead_time                             36275 non-null  int64
 9   arrival_year                          36275 non-null  int64
 10  arrival_month                         36275 non-null  int64
 11  arrival_date                          36275 non-null  int64
 12  market_segment_type                   36275 non-null  object
 13  repeated_guest                        36275 non-null  int64
 14  no_of_previous_cancellations          36275 non-null  int64
 15  no_of_previous_bookings_not_canceled  36275 non-null  int64
 16  avg_price_per_room                    36275 non-null  float64
 17  no_of_special_requests                36275 non-null  int64
 18  booking_status                        36275 non-null  object
dtypes: float64(1), int64(13), object(5)
memory usage: 5.3+ MB
```

- Booking_ID,type_of_meal_plan,room_type_reserved,market_segment_type,booking_status are of **object type**, avg_price_per_room is **float type**, while the rest columns are **numeric** in nature

Checking for missing values

	0
Booking_ID	0
no_of_adults	0
no_of_children	0
no_of_weekend_nights	0
no_of_week_nights	0
type_of_meal_plan	0
required_car_parking_space	0
room_type_reserved	0
lead_time	0
arrival_year	0
arrival_month	0
arrival_date	0
market_segment_type	0
repeated_guest	0
no_of_previous_cancellations	0
no_of_previous_bookings_not_canceled	0
avg_price_per_room	0
no_of_special_requests	0
booking_status	0

dtype: int64

- There are no null values in the dataset

Checking for Duplicate values

```
data.duplicated().sum()
```

```
np.int64(0)
```

- There are no duplicate values in the dataset

Dropping the columns with all unique values

The Booking_ID column was removed from the dataset as it contains unique values for each record and does not contribute to predicting booking cancellations.

	no_of_adults	no_of_children	no_of_weekend_nights	no_
0	2	0	1	
1	2	0	2	
2	1	0	2	
3	2	0	0	
4	2	0	1	

Statistical summary of the data

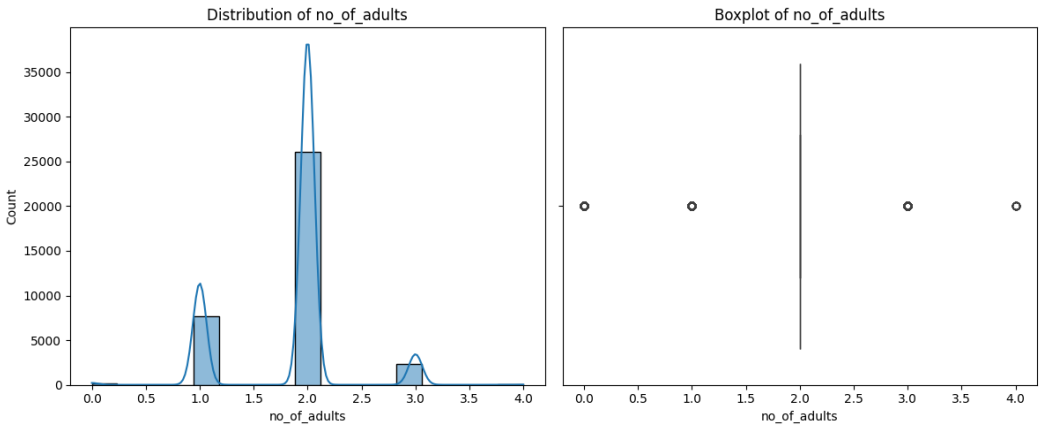
	count	mean	std	min	25%	50%	75%	max
no_of_adults	36275.00000	1.84496	0.51871	0.00000	2.00000	2.00000	2.00000	4.00000
no_of_children	36275.00000	0.10528	0.40265	0.00000	0.00000	0.00000	0.00000	10.00000
no_of_weekend_nights	36275.00000	0.81072	0.87064	0.00000	0.00000	1.00000	2.00000	7.00000
no_of_week_nights	36275.00000	2.20430	1.41090	0.00000	1.00000	2.00000	3.00000	17.00000
required_car_parking_space	36275.00000	0.03099	0.17328	0.00000	0.00000	0.00000	0.00000	1.00000
lead_time	36275.00000	85.23256	85.93082	0.00000	17.00000	57.00000	126.00000	443.00000
arrival_year	36275.00000	2017.82043	0.38384	2017.00000	2018.00000	2018.00000	2018.00000	2018.00000
arrival_month	36275.00000	7.42365	3.06989	1.00000	5.00000	8.00000	10.00000	12.00000
arrival_date	36275.00000	15.59700	8.74045	1.00000	8.00000	16.00000	23.00000	31.00000
repeated_guest	36275.00000	0.02564	0.15805	0.00000	0.00000	0.00000	0.00000	1.00000
no_of_previous_cancellations	36275.00000	0.02335	0.36833	0.00000	0.00000	0.00000	0.00000	13.00000
no_of_previous_bookings_not_canceled	36275.00000	0.15341	1.75417	0.00000	0.00000	0.00000	0.00000	58.00000
avg_price_per_room	36275.00000	103.42354	35.08942	0.00000	80.30000	99.45000	120.00000	540.00000
no_of_special_requests	36275.00000	0.61966	0.78624	0.00000	0.00000	0.00000	1.00000	5.00000

- The average number of adults per booking is 1.84, while the number of children is very low at 0.10, indicating that most bookings are made by adults only.
- Guests tend to stay longer on weekdays, with week nights averaging 2.20 compared to weekend nights averaging 0.81.
- The lead time has an average of 85.23 days, indicating that bookings are generally made well in advance.
- The proportion of repeated guests is very low, with an average of 0.026, indicating limited customer retention.
- Most customers have no history of previous cancellations, as seen from the low average of 0.02.
- The average price per room is 103.42, indicating variability in pricing across bookings.
- The number of special requests is generally low, with an average of 0.62.

Exploratory Data Analysis (EDA)

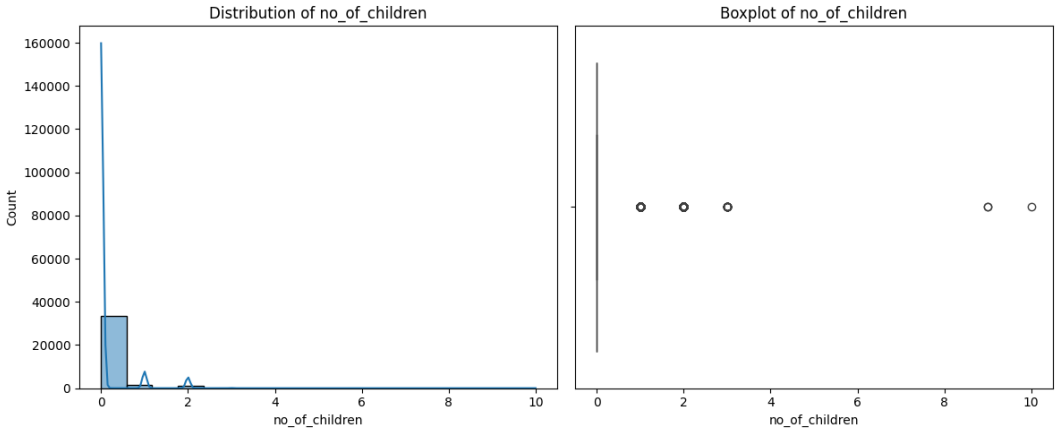
Univariate Analysis-Numerical Variables

Distribution of no_of_adults



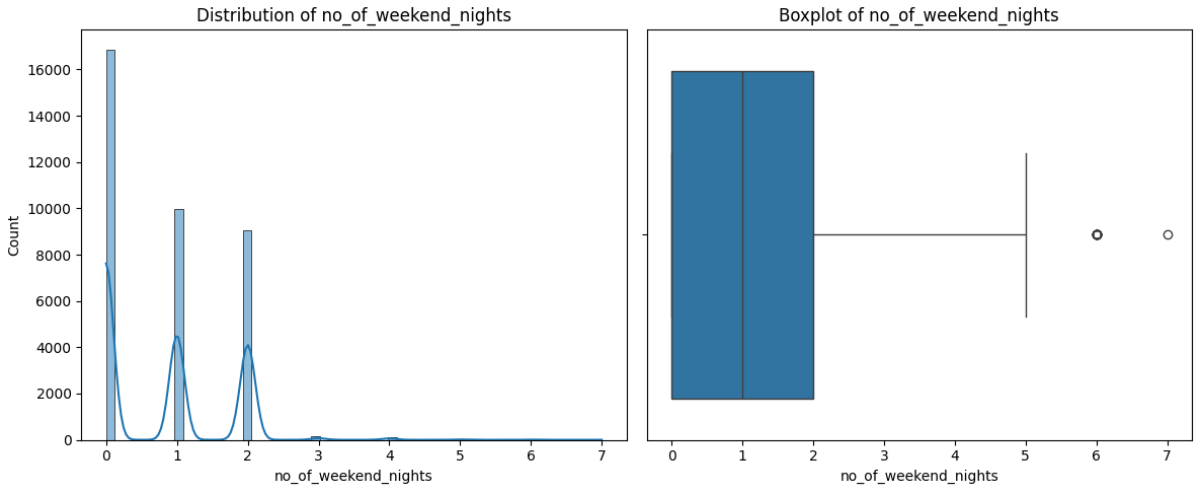
Most bookings are made for **2 adults**, indicating that couples or small groups form the majority of customers.

Distribution of no_of_children



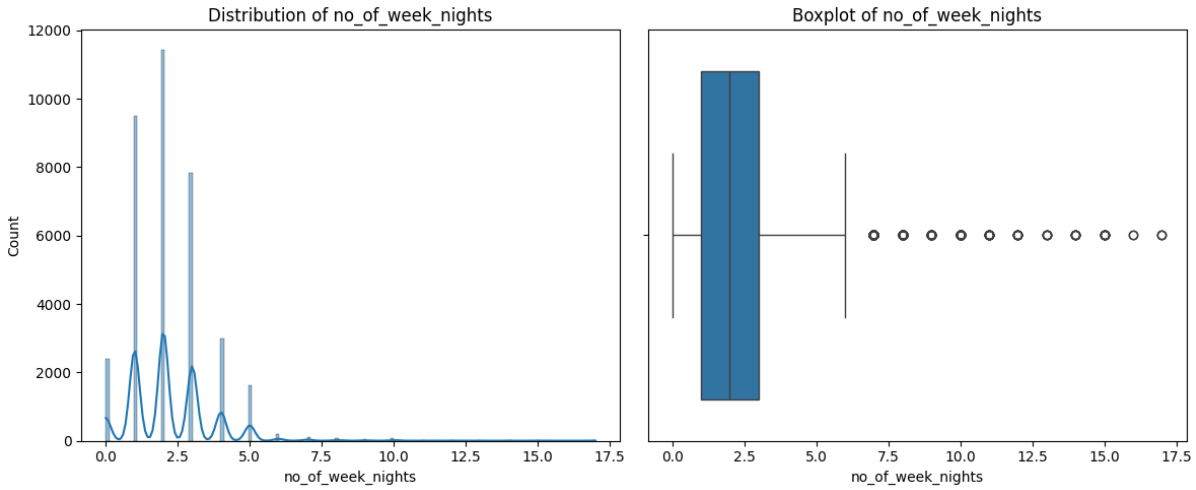
The number of children is mostly **0**, suggesting that bookings are primarily made by adults without children.

Distribution of no_of_weekend_nights



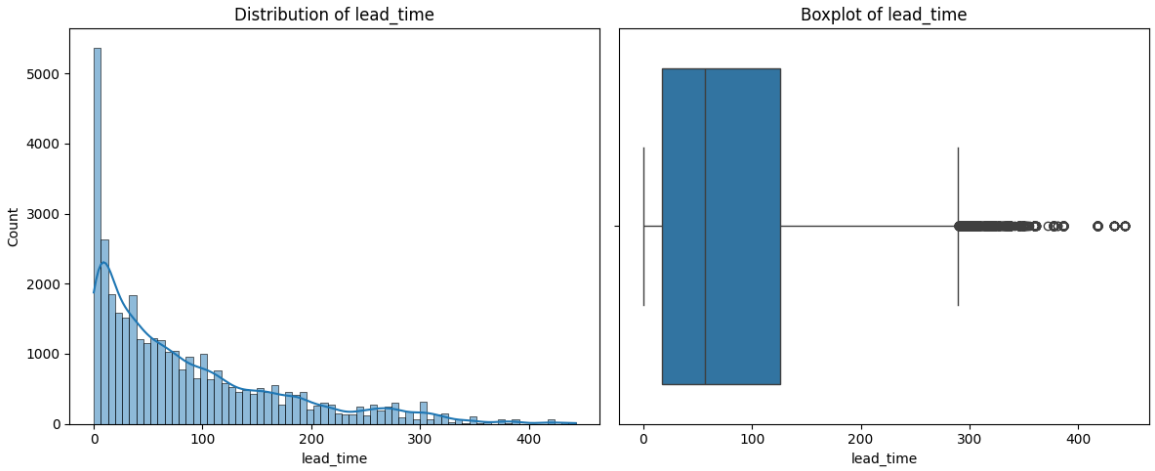
Most bookings have 0 or 1 weekend night stay There is a high concentration at 0, meaning many guests do not stay over weekends The distribution is right-skewed (few bookings with higher weekend nights) The boxplot shows outliers (5–7 nights), indicating very few long weekend stays

Distribution of no_of_week_nights



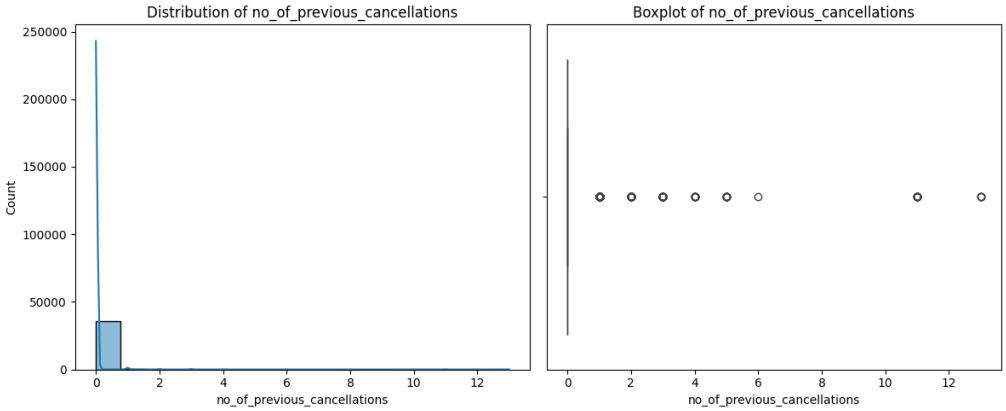
Most bookings have **1 to 3 weekday nights**, with a peak around **2 nights** .The distribution is **right-skewed**, meaning most stays are short, but a few extend to longerdurations .There are **significant outliers** (around 7–17 nights), indicating a small number of long-stay bookings The boxplot confirms that the majority of data lies within a **narrow range**, with extreme values on the higher side

Distribution of lead_time



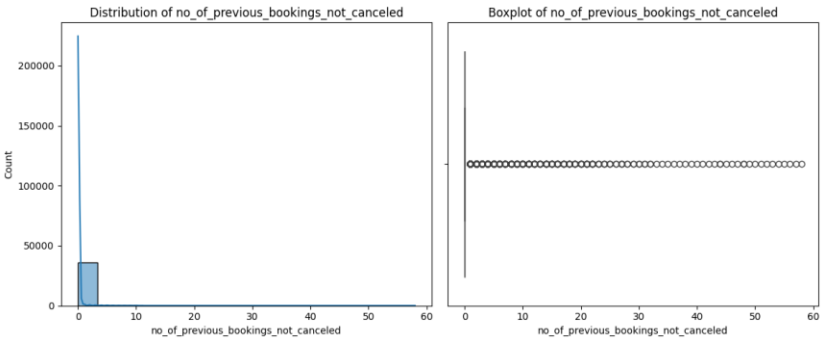
The distribution is **right-skewed**, indicating that while many bookings are made closer to the stay date, some are made **well in advance**, creating high variability.

Distribution of no_of_previous_cancellations



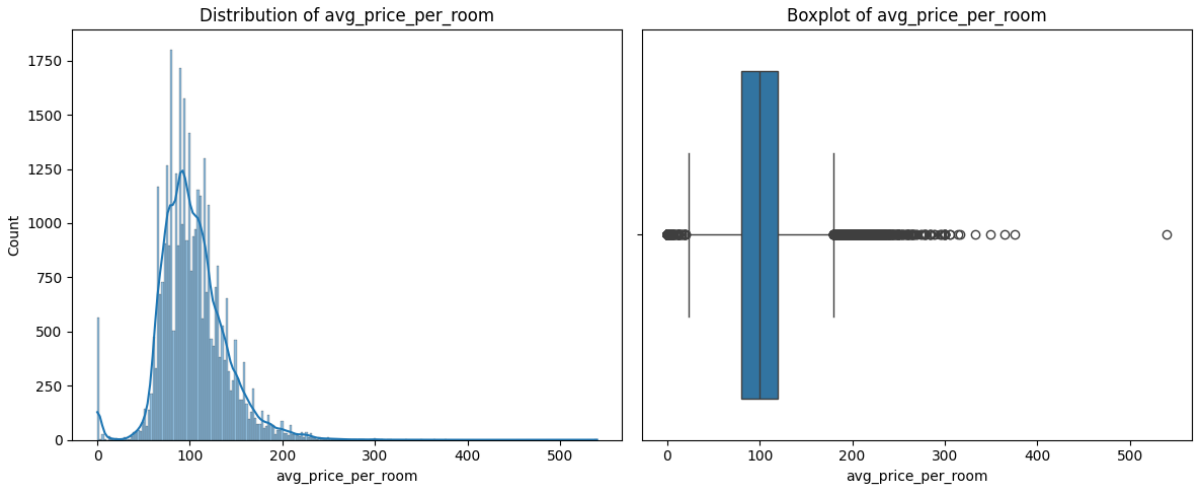
The majority of customers have **no previous cancellations**, indicating low historical cancellation behavior.

Distribution of no_of_previous_bookings_not_cancelled



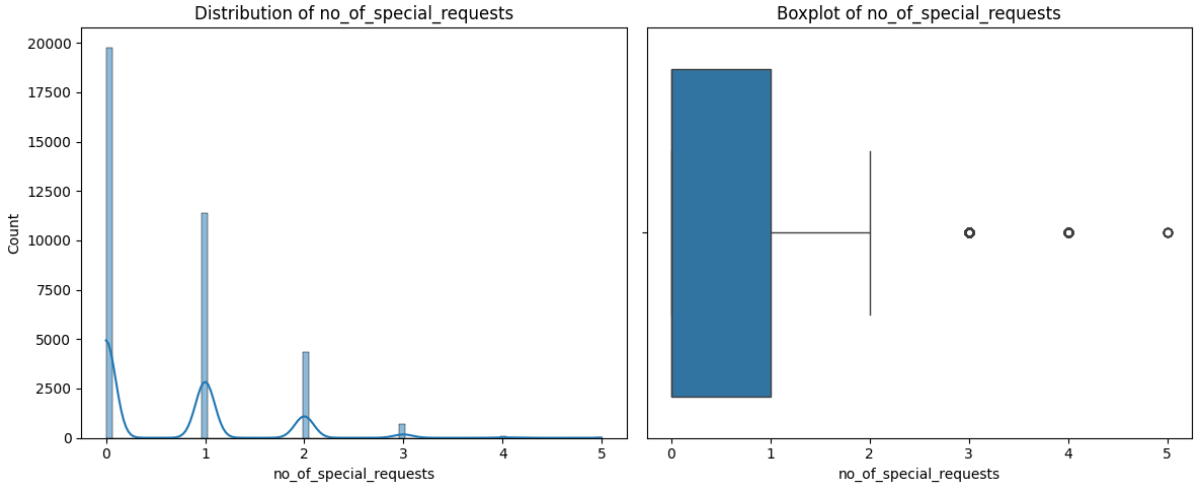
Most values are **0**, suggesting that a large portion of customers are either new or have limited booking history.

Distribution of avg_price_per_room



The distribution shows **variation in pricing**, indicating different customer segments, with some high-priced bookings acting as potential outliers.

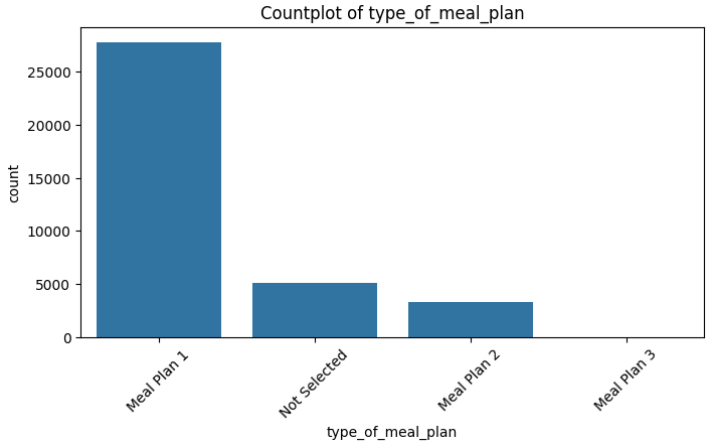
Distribution of no_of_special_requests



Most bookings have **0–1 special requests**, indicating that customers generally do not request many additional services

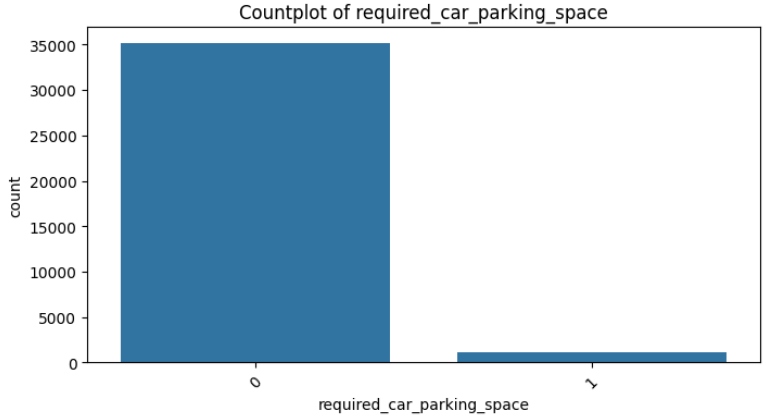
Univariate Analysis- Categorical Variables

Distribution of type_of_meal_plan



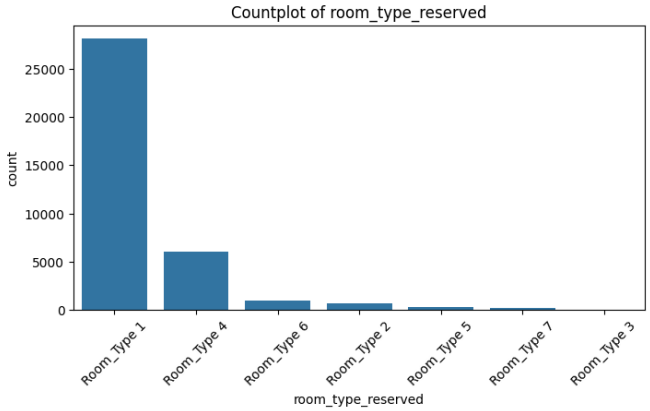
Meal plan 1 dominates the bookings, indicating a preferred choice among customers.

Distribution of required_car_parking_space



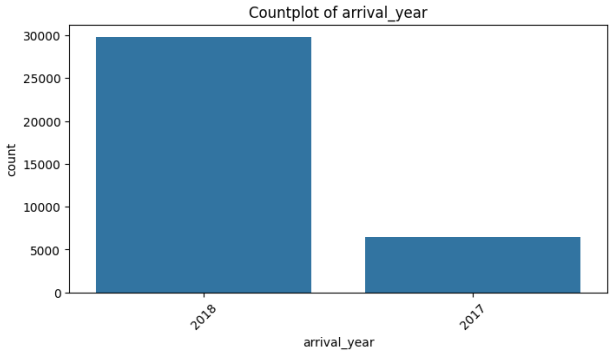
The majority of customers do not require parking, suggesting limited dependency on parking facilities**.**

Distribution of room_type_reserved



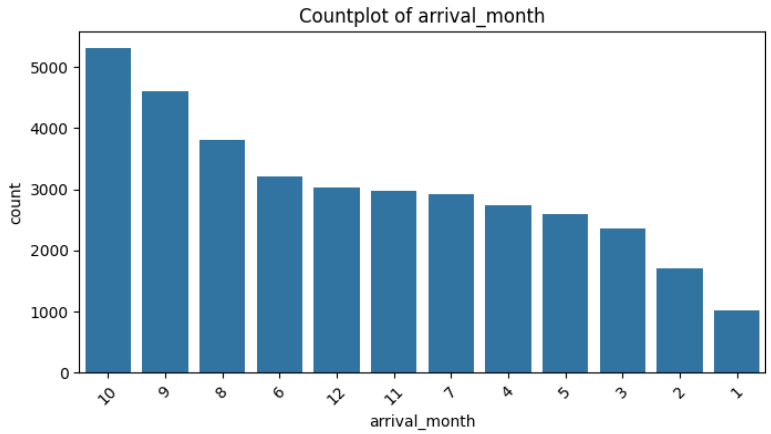
Room type 1 is more frequently booked followed by Room type 4 , indicating customer preference toward certain room categories**.**

Distribution of arrival_year



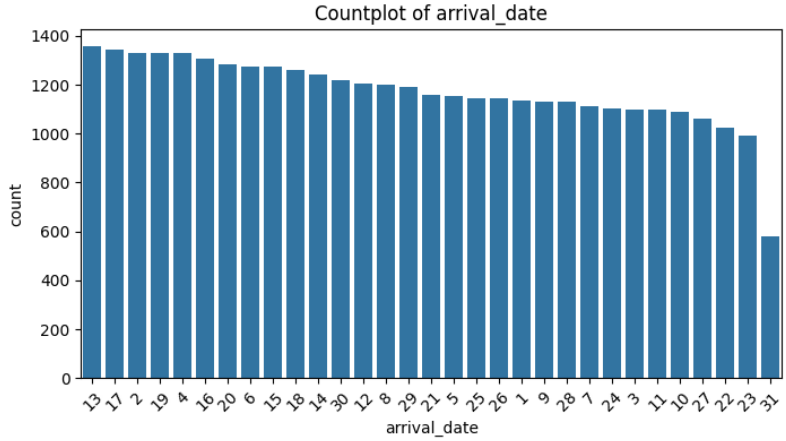
Most bookings are concentrated in a 2018year, indicating the dataset is time-bound.

Distribution of arrival_month



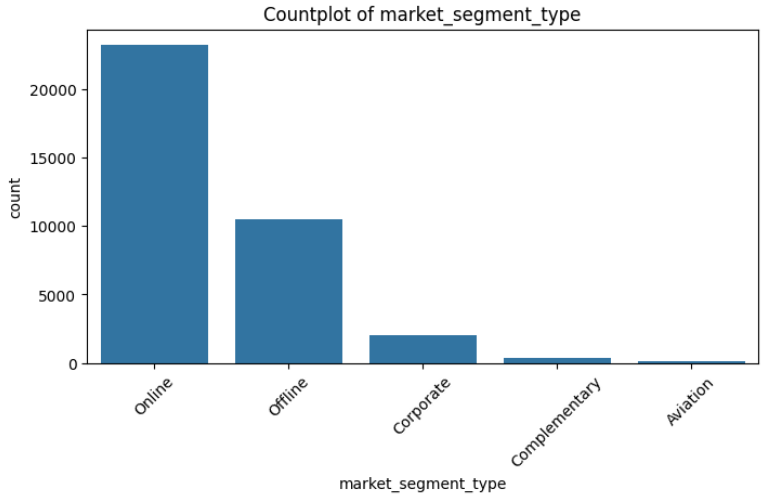
Bookings vary across months, showing seasonality in hotel demand.

Distribution of arrival_date



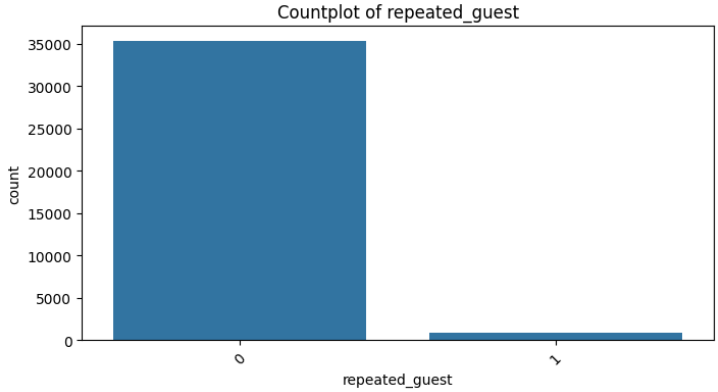
Bookings are fairly distributed across dates, with no extreme concentration on specific days.

Distribution of market_segment_type



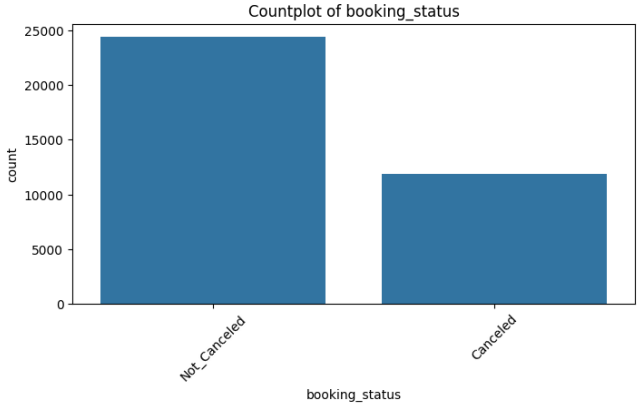
A particular segment which is online contributes the majority of bookings, highlighting the dominant booking channel.

Distribution of repeated_guest



Most customers are not repeat guests, indicating low customer retention.

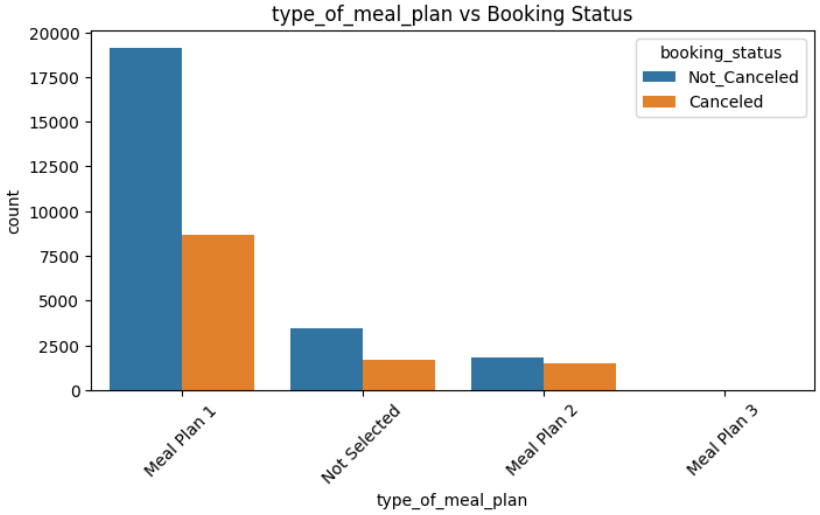
Distribution of booking_status



A significant portion of bookings are not cancelled, but cancellations still form a considerable share.

Bivariate analysis

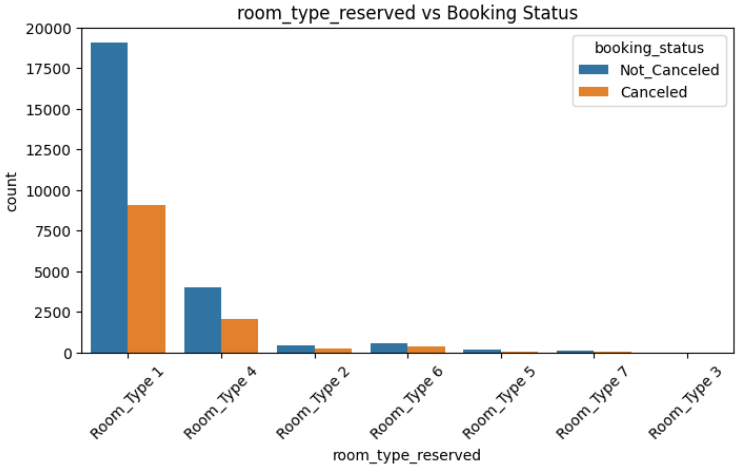
Comparing features with target (booking_status)- Canceled / Not_Canceled



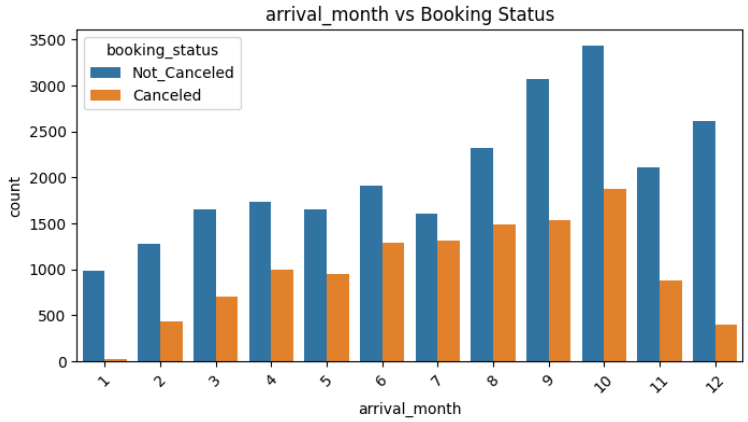
Meal plan 1 show higher cancellation rates, possibly due to lower commitment levels.



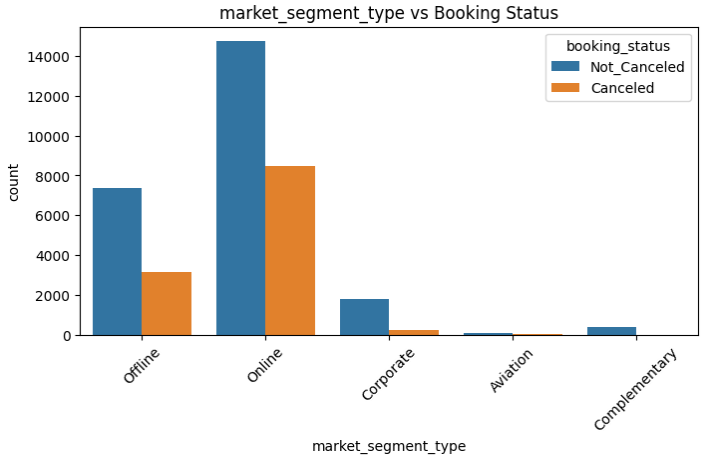
Customers who require parking are less likely to cancel, indicating higher commitment



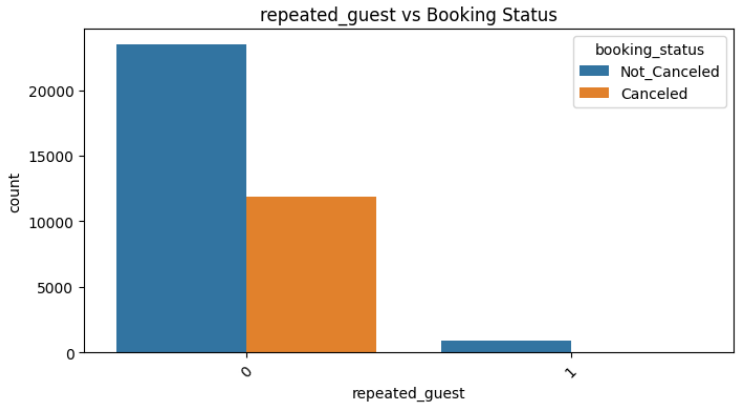
Room type 1 experience higher cancellations, possibly due to pricing or availability preferences.



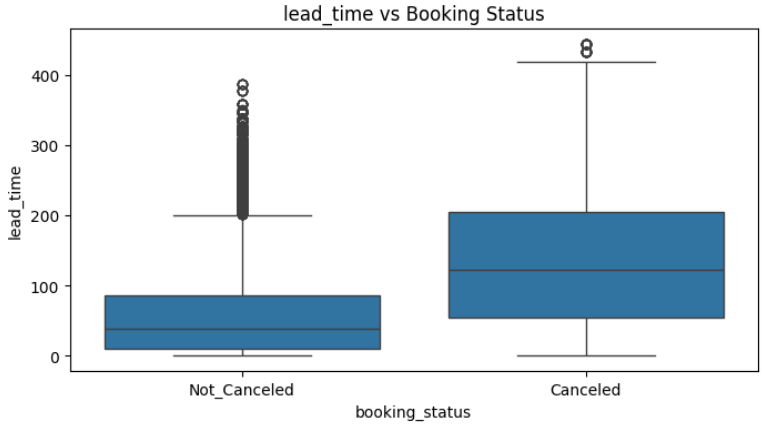
Cancellation rates vary across months, indicating seasonality in booking behaviour.



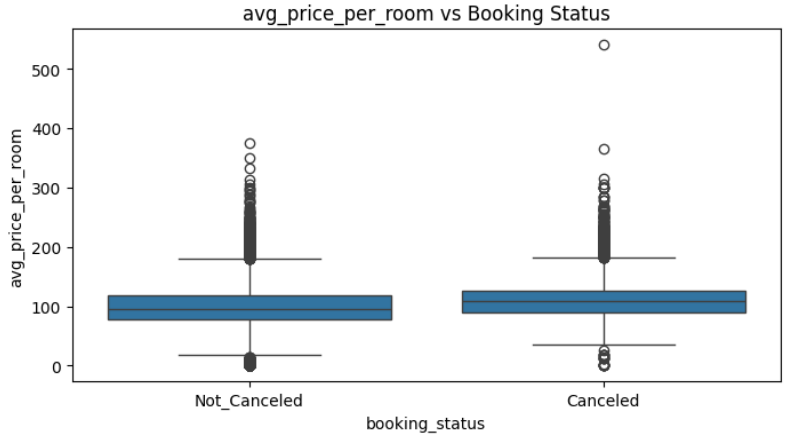
The online market segment shows a higher cancellation rate compared to other segments. This may be due to the flexibility and ease of cancellation in online booking



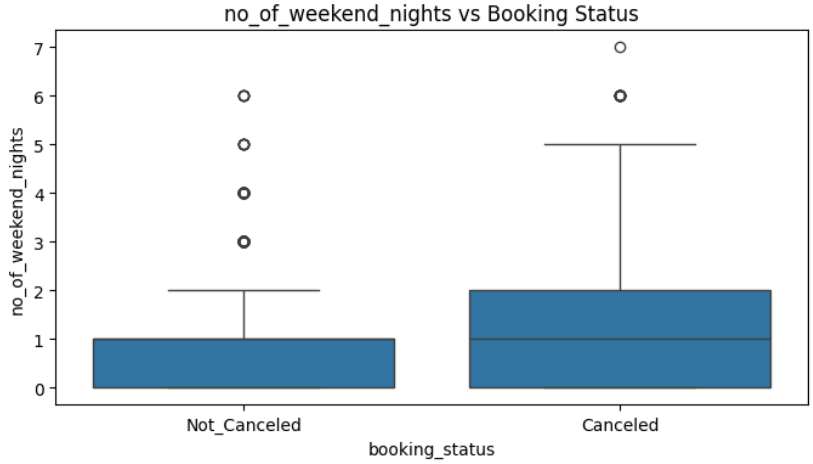
Repeated guests are far less likely to cancel compared to new customers



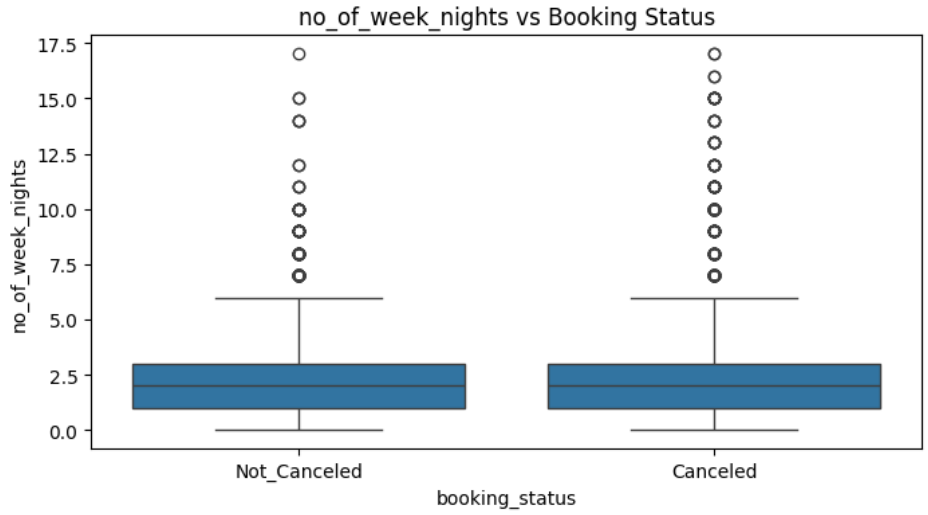
Bookings with higher lead time show a significantly higher probability of cancellation. Customers who book well in advance are more likely to change or cancel their plans.



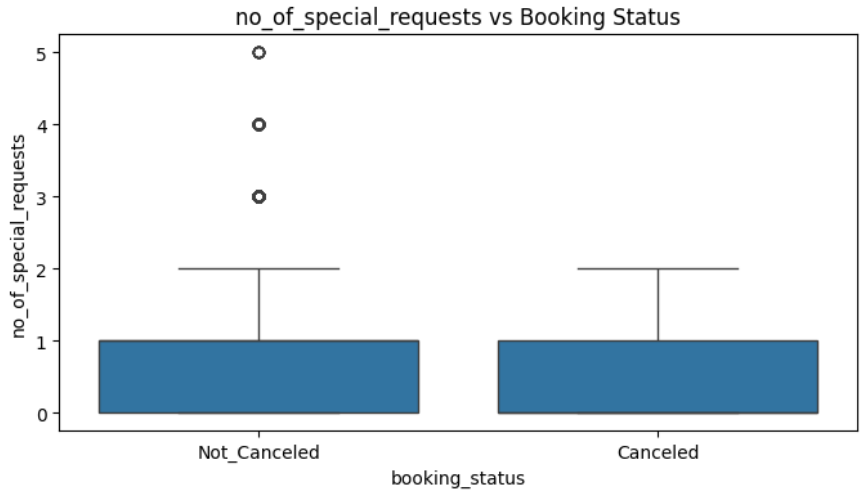
Bookings with higher room prices tend to have slightly higher cancellation rates. Price-sensitive customers may cancel if they find better alternatives



Short weekend stays (0–1 nights) show higher cancellation rates, while longer stays are relatively stable.

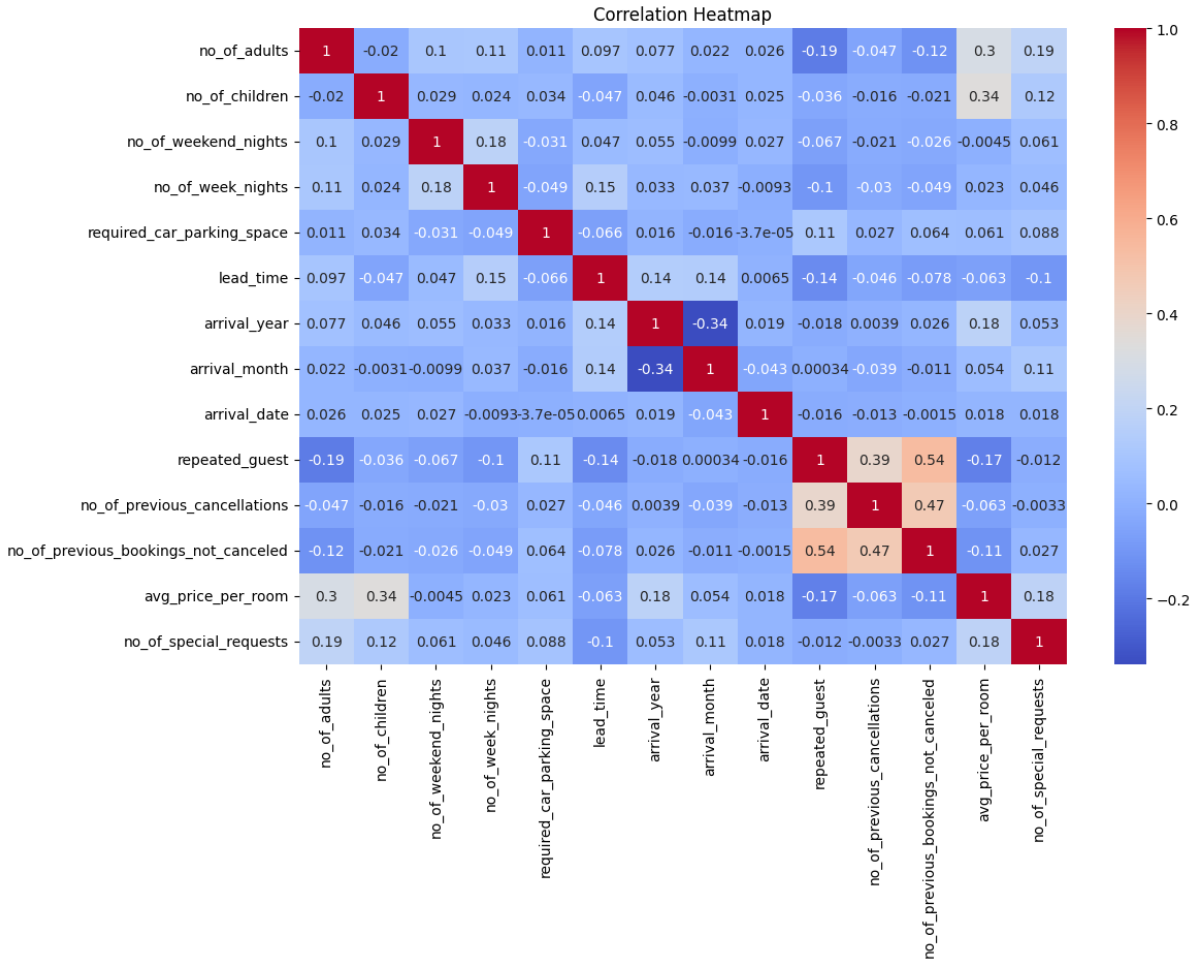


Bookings with longer weekday stays show slightly lower cancellation rates. Longer stays indicate stronger intent to complete the booking.



Customers with more special requests are less likely to cancel. This suggests that engaged customers with specific needs are more committed to their bookings.

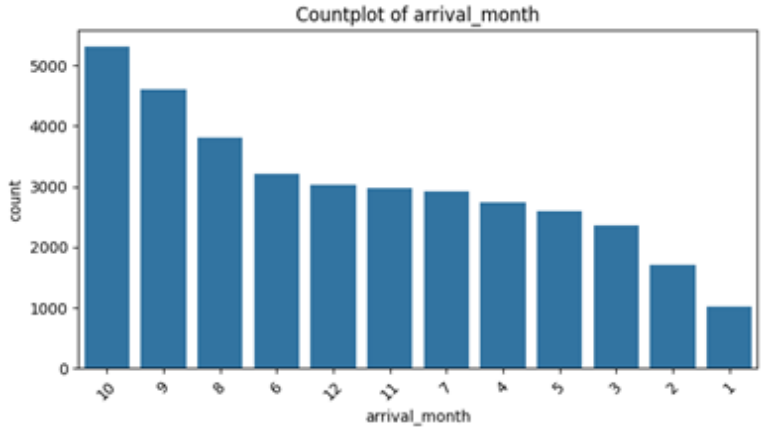
Correlation



The correlation heatmap shows that most variables have weak correlations, indicating no significant multicollinearity in the dataset. However, moderate relationships are observed among customer history variables (like repeated guests and previous bookings), while pricing is slightly influenced by the number of adults and children.”

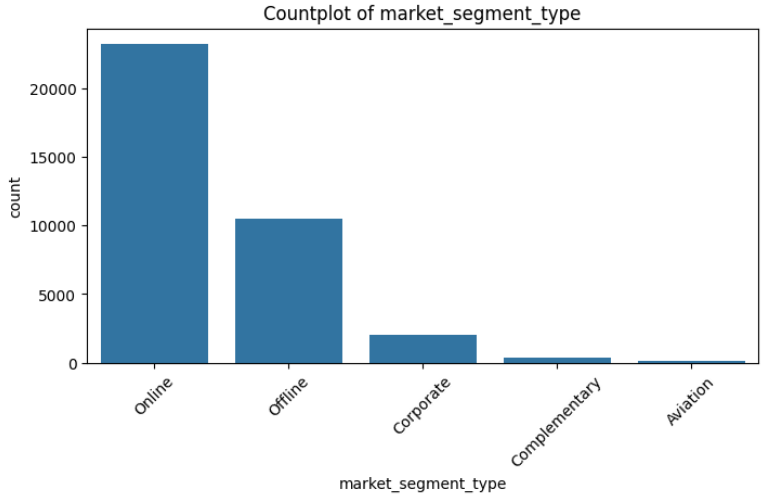
EDA Questions:

1. What are the busiest months in the hotel?



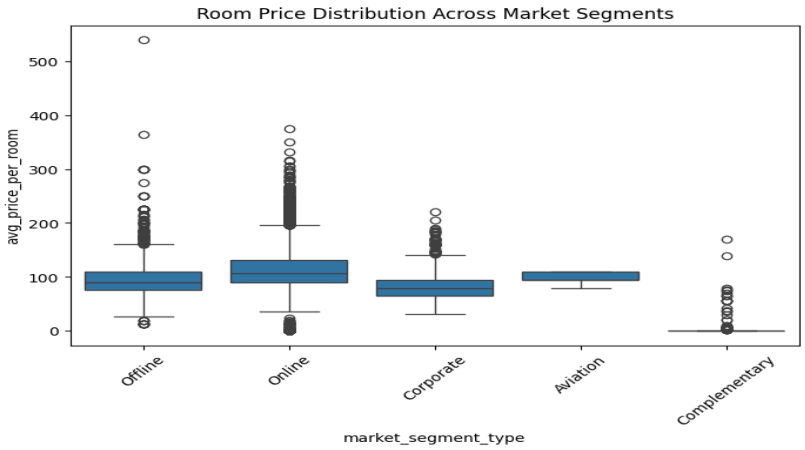
- The hotel experiences the highest number of bookings in the months of October (10) and September (9), followed by August (8).
- In contrast, the lowest number of bookings are observed in January (1) and February (2), suggesting reduced demand during the early months of the year.

2. Which market segment do most of the guests come from?



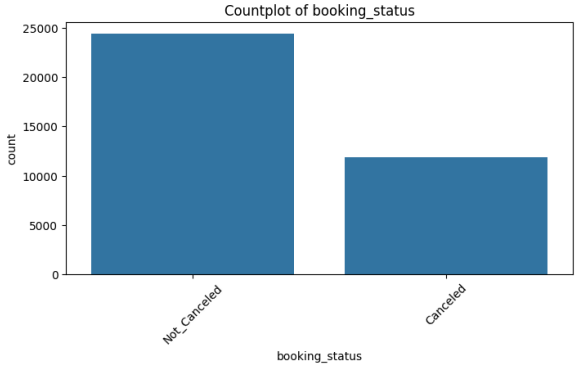
- The majority of guests come from the Online market segment, which has the highest number of bookings compared to other segments.
- This is followed by the Offline segment, while Corporate, Complementary, and Aviation segments contribute significantly fewer booking

3. Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?



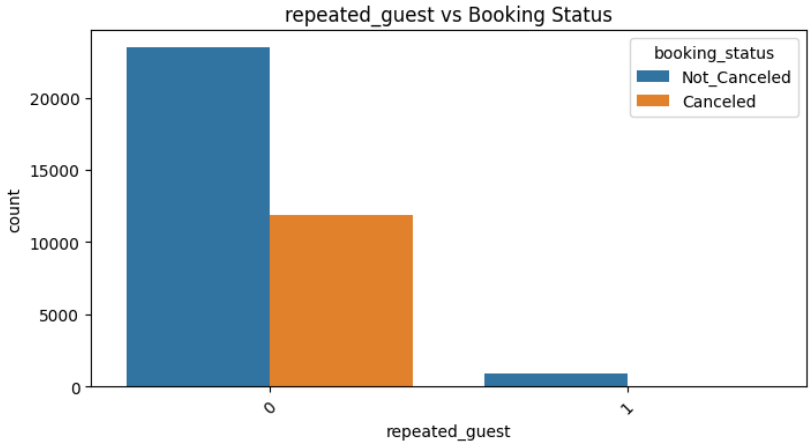
- The variation in room prices across different market segments indicates that hotel rates are dynamic in nature.
- Segments such as Online show higher and more variable prices, likely due to higher demand and flexible pricing strategies.
- In contrast, segments like Corporate have more stable and lower prices, reflecting negotiated or fixed rates. Additionally, Complementary bookings have minimal or zero pricing.
- This demonstrates that hotel pricing is influenced by demand patterns and customer demographics across different segments

4. What percentage of bookings are cancelled?



- Approximately 33% of the bookings are cancelled, while around 67% are not cancelled.
- This indicates that a significant proportion of bookings result in cancellations, which can have a considerable impact on the hotel's revenue and operational planning
- The relatively high cancellation rate highlights the need for predictive models to identify potential cancellations in advance

5. Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?



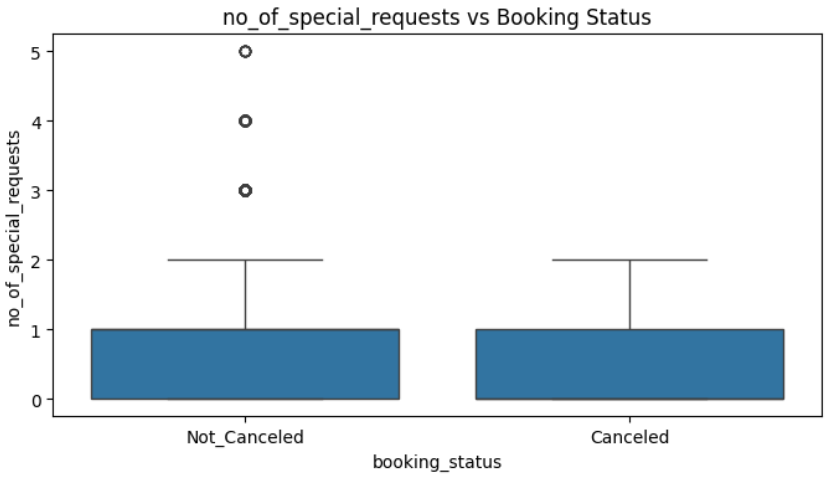
0 = Non-repeated guests

- High cancellations (~12,000)
- High non-cancellations (~23,000)

1 = Repeated guests

- Almost no cancellations
- Mostly not cancelled
- The percentage of cancellations among repeating guests is extremely low compared to non-repeating guests. Most repeated guests complete their bookings without cancellation, indicating strong customer loyalty and commitment.
- In contrast, cancellations are significantly higher among non-repeated guests.

6. Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?



- Customers with a higher number of special requests are less likely to cancel their bookings.
- This indicates that guests who are more engaged and have specific requirements tend to be more committed to their stay. In contrast, bookings with fewer or no special requests show a higher likelihood of cancellation.

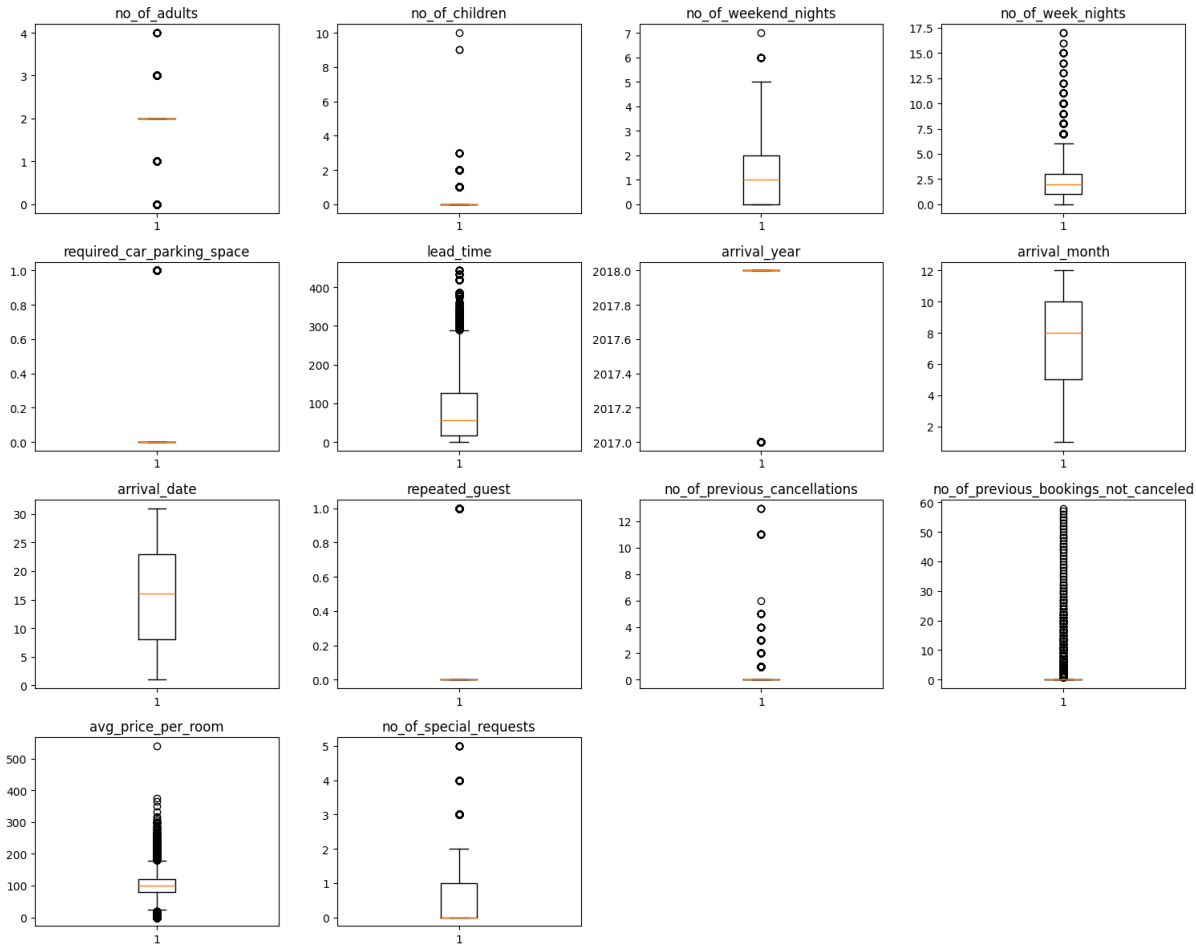
Data Preprocessing

1. Missing Value Treatment

	0
Booking_ID	0
no_of_adults	0
no_of_children	0
no_of_weekend_nights	0
no_of_week_nights	0
type_of_meal_plan	0
required_car_parking_space	0
room_type_reserved	0
lead_time	0
arrival_year	0
arrival_month	0
arrival_date	0
market_segment_type	0
repeated_guest	0
no_of_previous_cancellations	0
no_of_previous_bookings_not_canceled	0
avg_price_per_room	0
no_of_special_requests	0
booking_status	0

- The dataset was checked for missing values, and it was observed that there are no missing values present in any of the variables**.**

2.Outlier Detection



- Lead Time
  Shows a large number of extreme values (very high booking days), indicating that some customers book far in advance.

- Average Price per Room
  Contains several high-value outliers, representing premium or luxury bookings.

- Number of Week Nights
  Most stays are short, but a few bookings extend to very long durations, visible as outliers.

- Previous Bookings (Not Cancelled)
  Highly skewed with extreme values, indicating that a small group of customers has frequent booking history.

- Number of Children & Special Requests
  Mostly concentrated at lower values, with a few extreme cases.

- Binary Variables (like repeated_guest, required_car_parking_space) Show apparent “outliers,” but these are not actual outliers—they are just due to categorical nature (0/1 values).

- Although several variables exhibit outliers, these values represent real-world customer behaviour such as long stays, early bookings, and premium pricing. Therefore, outliers were retained to preserve meaningful information for predictive modelling

3.Feature Engineering

	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_
0	2	0	1	2	
1	2	0	2	3	
2	1	0	2	1	
3	2	0	0	2	
4	2	0	1	1	

type_of_meal_plan_Meal Plan 2	type_of_meal_plan_Meal Plan 3	type_of_meal_plan_Not Selected	room_type_reserved_Room_Type 2	room_type_reserved_Room_Type 3	room_type_reserved_Room_Type 4	room_type_reserved_Room_Type 5	room_type_reserved_Room_Type 6	room_
False	False	False	False	False	False	False	False	
False	False	True	False	False	False	False	False	
False	False	False	False	False	False	False	False	
False	False	False	False	False	False	False	False	
False	False	True	False	False	False	False	False	

- The ‘Booking_ID’ column was removed as it is a unique identifier and does not contribute to prediction.
- Categorical variables were encoded into numerical format using one-hot encoding to make them suitable for machine learning model

4. Data Scaling

```
# AFTER scaling
X[num_cols].head()
```

	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	required_car_parking_space	lead_time	arrival_year	arrival_month	arrival_date	repeated_guest	no_of_previous_cancellations	no_of_previous_bookings_not_canceled
0	0.29889	-0.26147	0.21740	-0.14480	-0.17882	1.61490	-2.13747	0.83924	-1.55566	-0.16221	-0.06339	-0.08746
1	0.29889	-0.26147	1.36599	0.56397	-0.17882	-0.93370	0.46784	1.16499	-1.09801	-0.16221	-0.06339	-0.08746
2	-1.62898	-0.26147	1.36599	-0.85358	-0.17882	-0.98025	0.46784	-1.76675	1.41906	-0.16221	-0.06339	-0.08746
3	0.29889	-0.26147	-0.93119	-0.14480	-0.17882	1.46361	0.46784	-0.78950	0.50376	-0.16221	-0.06339	-0.08746
4	0.29889	-0.26147	0.21740	-0.85358	-0.17882	-0.43329	0.46784	-1.11525	-0.52595	-0.16221	-0.06339	-0.08746

canceled	avg_price_per_room	no_of_special_requests
-0.08746	-1.09503	-0.78814
-0.08746	0.09281	0.48376
-0.08746	-1.23753	-0.78814
-0.08746	-0.09757	-0.78814
-0.08746	-0.25431	-0.78814

Feature scaling was performed using StandardScaler to normalize numerical variables. This ensures that all features are on a similar scale, improving model performance, especially for distance-based algorithms.

5. Train-Test Split

```
Shape of Training set :  (29020, 27)
Shape of test set :  (7255, 27)
Percentage of classes in training set:
booking_status_Not_Canceled
1   0.67371
0   0.32629
Name: proportion, dtype: float64
Percentage of classes in test set:
booking_status_Not_Canceled
1   0.66699
0   0.33301
Name: proportion, dtype: float64
```

1 → Not Cancelled

0 → Cancelled

- The dataset was split into training and testing sets using an 80:20 ratio. The training set consists of 29,020 observations, while the test set contains 7,255 observations.
- The class distribution remains consistent across both sets, with approximately 67% of bookings not cancelled and 33% cancelled. This indicates that the data is well-balanced across training and testing sets, ensuring reliable model evaluation

Model Building

1. Logistic Regression

- Since the objective is to predict booking cancellations, recall is chosen as the primary evaluation metric. This is because correctly identifying cancellations is more important for the business to minimize revenue loss. However, accuracy, precision, and F1-score are also evaluated for overall model performance

Logit Regression Results

Dep. Variable:	booking_status_Not_Canceled	**No. Observations:**	29020
Model:	Logit	**Df Residuals:**	28992
Method:	MLE	**Df Model:**	27
Date:	Sat, 25 Apr 2026	**Pseudo R-squ.:**	0.3292
Time:	08:18:40	**Log-Likelihood:**	-12294.
converged:	False	**LL-Null:**	-18327.
Covariance Type:	nonrobust	**LLR p-value:**	0.000

	coef	std err	z	P>|z|	[0.025	0.975]
const	834.7475	112.575	7.415	0.000	614.105	1055.390
no_of_adults	-0.1116	0.035	-3.164	0.002	-0.181	-0.042
no_of_children	-0.1188	0.055	-2.156	0.031	-0.227	-0.011
no_of_weekend_nights	-0.1262	0.019	-6.787	0.000	-0.163	-0.090
no_of_week_nights	-0.0402	0.012	-3.496	0.000	-0.063	-0.018
required_car_parking_space	1.6570	0.132	12.572	0.000	1.399	1.915
lead_time	-0.0156	0.000	-62.545	0.000	-0.016	-0.015
arrival_year	-0.4123	0.056	-7.391	0.000	-0.522	-0.303
arrival_month	0.0434	0.006	7.121	0.000	0.031	0.055
arrival_date	-0.0014	0.002	-0.744	0.457	-0.005	0.002
repeated_guest	2.2420	0.521	4.305	0.000	1.221	3.263
no_of_previous_cancellations	-0.2366	0.070	-3.400	0.001	-0.373	-0.100
no_of_previous_bookings_not_canceled	0.0725	0.082	0.880	0.379	-0.089	0.234
avg_price_per_room	-0.0188	0.001	-27.238	0.000	-0.020	-0.017
no_of_special_requests	1.4956	0.028	52.659	0.000	1.440	1.551
type_of_meal_plan_Meal Plan 2	-0.1935	0.062	-3.113	0.002	-0.315	-0.072
type_of_meal_plan_Meal Plan 3	-18.5021	7228.635	-0.003	0.998	-1.42e+04	1.41e+04
type_of_meal_plan_Not Selected	-0.2292	0.050	-4.613	0.000	-0.327	-0.132
room_type_reserved_Room_Type 2	0.4074	0.123	3.315	0.001	0.167	0.648
room_type_reserved_Room_Type 3	0.1610	1.409	0.114	0.909	-2.601	2.923
room_type_reserved_Room_Type 4	0.2852	0.050	5.715	0.000	0.187	0.383
room_type_reserved_Room_Type 5	0.6585	0.191	3.441	0.001	0.283	1.034
room_type_reserved_Room_Type 6	0.9453	0.140	6.764	0.000	0.671	1.219
room_type_reserved_Room_Type 7	1.3770	0.283	4.862	0.000	0.822	1.932
market_segment_type_Complementary	29.0395	7318.643	0.004	0.997	-1.43e+04	1.44e+04
market_segment_type_Corporate	0.9401	0.259	3.627	0.000	0.432	1.448
market_segment_type_Offline	1.8790	0.249	7.551	0.000	1.391	2.367
market_segment_type_Online	0.0894	0.246	0.363	0.716	-0.393	0.572

Terminal

- The model is statistically significant with a Pseudo R^2 of 0.329, indicating moderate explanatory power.
- Key factors influencing booking cancellations include lead time, average price per room, number of weekend nights, and customer engagement variables such as repeated guests and special requests.
- Higher lead time and room prices increase the likelihood of cancellation, while repeated guests and customers with special requests are less likely to cancel. However, the model did not fully converge, indicating that the results should be interpreted with caution

  .

  Evaluation

```
Accuracy: 0.80868366643694
Precision: 0.8323809523809523
Recall: 0.8958589585895859
F1 Score: 0.8629541864139021
```

- The Logistic Regression model achieved an accuracy of 80%, with a precision of 83% and recall of 89.46%.
- The high recall indicates that the model is effective in identifying bookings that are not cancelled.
- The F1-score of 86% reflects a good balance between precision and recall, suggesting overall strong model performance

2.Decision Tree

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=1)
```

Model Evaluation Criterion

In the context of hotel booking cancellations, the model can make two types of errors:

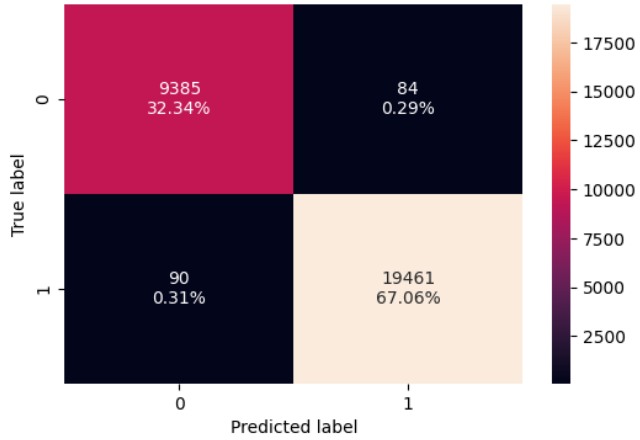
- False Negative (FN): Predicting that a booking will not be cancelled, but in reality, it gets cancelled.
- False Positive (FP): Predicting that a booking will be cancelled, but in reality, it is not cancelled.

Among these, False Negatives are more critical.
If the model predicts that a booking will not be cancelled but it actually gets cancelled, the hotel may face revenue loss due to unsold rooms, last-minute price drops, and inefficient resource planning.

On the other hand, if the model predicts a cancellation incorrectly (False Positive), the hotel may take preventive actions such as overbooking or promotional offers, which generally incur lower costs compared to actual revenue loss.

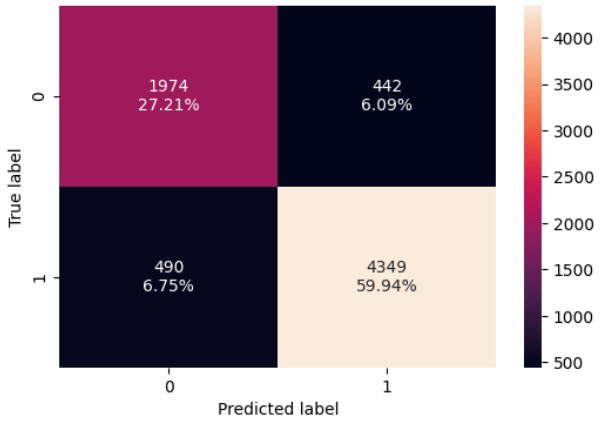
Therefore, to minimize business losses, the model should aim to maximize recall for the cancelled bookings. Higher recall ensures that most actual cancellations are correctly identified in advance

Confusion matrix for model0 on X_train and y_train



	Accuracy	Recall	Precision	F1
0	0.99400	0.99540	0.99570	0.99555

Confusion matrix X_test and y_test.

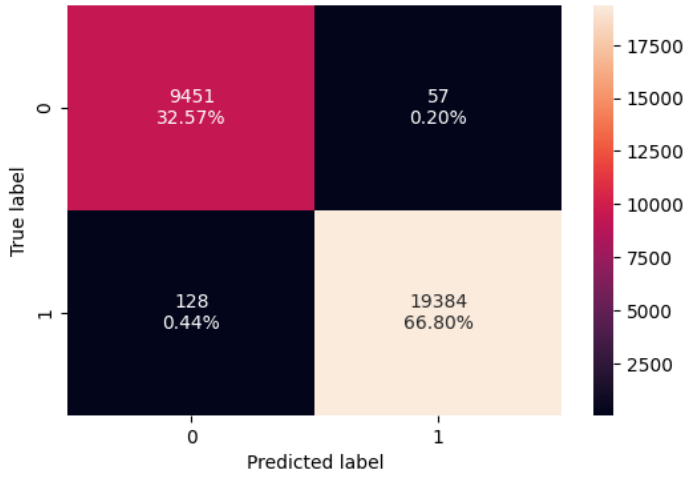


Test performance of model0 on X_test and y_test

	Accuracy	Recall	Precision	F1
)	0.87154	0.89874	0.90774	0.90322

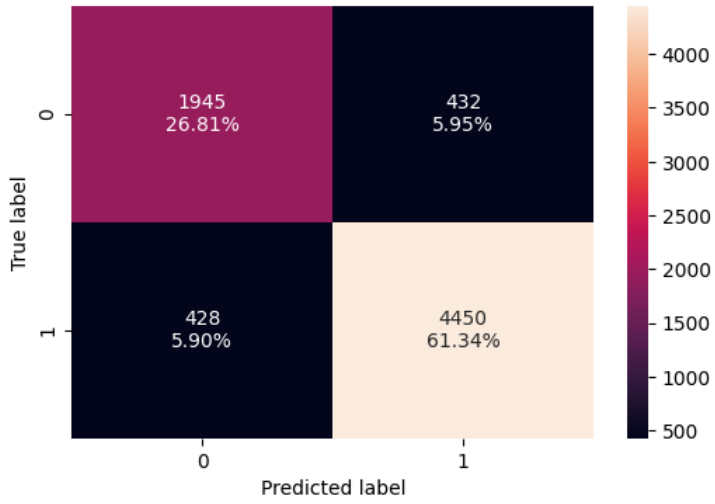
Decision Tree (with class_weights)

confusion_matrix_sklearn(model, X_train, y_train)



	Accuracy	Recall	Precision	F1
0	0.99363	0.99344	0.99707	0.99525

Confusion matrix for model on X_test and y_test.



	Accuracy	Recall	Precision	F1
0	0.88146	0.91226	0.91151	0.91189

- The model demonstrates excellent performance on the training data, achieving an accuracy of 99.40% and F1-score of 99.55%. However, on the test data, the performance drops to an accuracy of 87.15% and F1-score of 90.32%. This difference indicates that the model is overfitting the training data and may not generalize perfectly to unseen data.

- Despite this, the model maintains a high recall of 89.87% on the test set, which is crucial for identifying booking cancellations. Therefore, while the model performs well, further tuning is done to reduce overfitting

Model Performance

- The performance of both models was evaluated using accuracy, precision, recall, and F1-score. The Logistic Regression model achieved an accuracy of 80.65%, with a precision of 82.88% and recall of 89.46%, indicating strong overall performance and a good balance between precision and recall.
- Recall is considered the most important metric in this problem, as identifying potential cancellations is critical for minimizing revenue loss. A higher recall ensures that most actual cancellations are correctly identified by the model.
- The Decision Tree model was also evaluated and compared against Logistic Regression. While Decision Trees are capable of capturing non-linear relationships and may provide better interpretability, they are also prone to overfitting. Based on the comparison of evaluation metrics, the model with higher recall and balanced F1-score is considered more suitable for the problem.

Model Performance Improvement
- Tune the following models to improve performance
1. Logistic Regression tuning

Remove High p-value Variables

Logit Regression Results

Dep. Variable:	booking_status_Not_Canceled	**No. Observations:**	29020
Model:	Logit	**Df Residuals:**	28996
Method:	MLE	**Df Model:**	23
Date:	Sun, 26 Apr 2026	**Pseudo R-squ.:**	0.3283
Time:	06:17:51	**Log-Likelihood:**	-12309.
converged:	True	**LL-Null:**	-18327.
Covariance Type:	nonrobust	**LLR p-value:**	0.000

	coef	std err	z	P>|z|	[0.025	0.975]
const	821.4264	112.265	7.317	0.000	601.391	1041.462
no_of_adults	-0.1124	0.035	-3.214	0.001	-0.181	-0.044
no_of_children	-0.1165	0.055	-2.114	0.035	-0.224	-0.008
no_of_weekend_nights	-0.1281	0.019	-6.899	0.000	-0.164	-0.092
no_of_week_nights	-0.0417	0.011	-3.632	0.000	-0.064	-0.019
required_car_parking_space	1.6611	0.132	12.602	0.000	1.403	1.919
lead_time	-0.0156	0.000	-62.929	0.000	-0.016	-0.015
arrival_year	-0.4056	0.056	-7.291	0.000	-0.515	-0.297
arrival_month	0.0448	0.006	7.384	0.000	0.033	0.057
repeated_guest	2.2584	0.520	4.347	0.000	1.240	3.277
no_of_previous_cancellations	-0.2381	0.070	-3.420	0.001	-0.375	-0.102
no_of_previous_bookings_not_canceled	0.0718	0.082	0.874	0.382	-0.089	0.233
avg_price_per_room	-0.0192	0.001	-28.345	0.000	-0.021	-0.018
no_of_special_requests	1.4945	0.028	52.696	0.000	1.439	1.550
type_of_meal_plan_Meal Plan 2	-0.1819	0.062	-2.931	0.003	-0.304	-0.060
type_of_meal_plan_Not Selected	-0.2402	0.050	-4.848	0.000	-0.337	-0.143
room_type_reserved_Room_Type 2	0.3977	0.123	3.240	0.001	0.157	0.638
room_type_reserved_Room_Type 3	0.2629	1.351	0.195	0.846	-2.384	2.910
room_type_reserved_Room_Type 4	0.2901	0.050	5.828	0.000	0.193	0.388
room_type_reserved_Room_Type 5	0.6758	0.191	3.540	0.000	0.302	1.050
room_type_reserved_Room_Type 6	0.9688	0.140	6.940	0.000	0.695	1.242
room_type_reserved_Room_Type 7	1.4177	0.283	5.012	0.000	0.863	1.972
market_segment_type_Corporate	0.8352	0.097	8.648	0.000	0.646	1.024
market_segment_type_Offline	1.7778	0.049	36.634	0.000	1.683	1.873

Logit Regression Results

Dep. Variable:	booking_status_Not_Canceled	**No. Observations:**	29020
Model:	Logit	**Df Residuals:**	28998
Method:	MLE	**Df Model:**	21
Date:	Sun, 26 Apr 2026	**Pseudo R-squ.:**	0.3283
Time:	06:51:42	**Log-Likelihood:**	-12310.
converged:	True	**LL-Null:**	-18327.
Covariance Type:	nonrobust	**LLR p-value:**	0.000

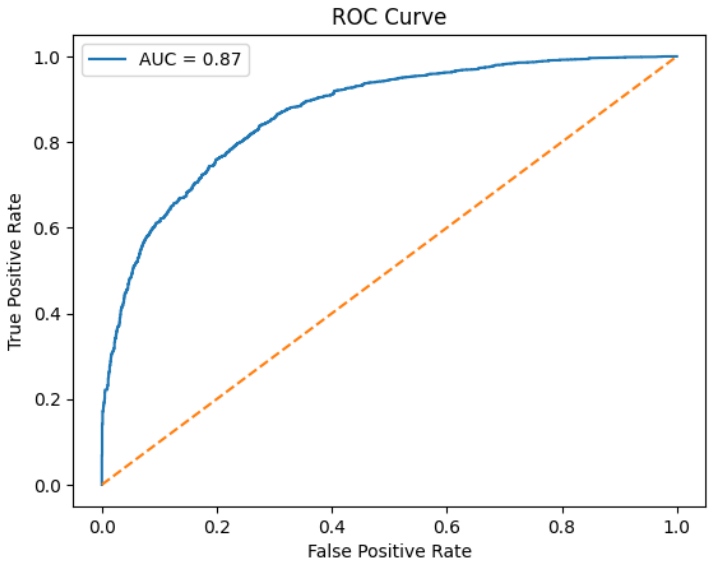
	coef	std err	z	P>|z|	[0.025	0.975]
const	820.2437	112.270	7.306	0.000	600.199	1040.288
no_of_adults	-0.1123	0.035	-3.212	0.001	-0.181	-0.044
no_of_children	-0.1165	0.055	-2.114	0.034	-0.224	-0.009
no_of_weekend_nights	-0.1280	0.019	-6.897	0.000	-0.164	-0.092
no_of_week_nights	-0.0417	0.011	-3.631	0.000	-0.064	-0.019
required_car_parking_space	1.6611	0.132	12.602	0.000	1.403	1.919
lead_time	-0.0156	0.000	-62.951	0.000	-0.016	-0.015
arrival_year	-0.4050	0.056	-7.280	0.000	-0.514	-0.296
arrival_month	0.0448	0.006	7.387	0.000	0.033	0.057
repeated_guest	2.4806	0.481	5.160	0.000	1.538	3.423
no_of_previous_cancellations	-0.2251	0.067	-3.375	0.001	-0.356	-0.094
avg_price_per_room	-0.0192	0.001	-28.353	0.000	-0.021	-0.018
no_of_special_requests	1.4948	0.028	52.709	0.000	1.439	1.550
type_of_meal_plan_Meal Plan 2	-0.1816	0.062	-2.925	0.003	-0.303	-0.060
type_of_meal_plan_Not Selected	-0.2403	0.050	-4.852	0.000	-0.337	-0.143
room_type_reserved_Room_Type 2	0.3978	0.123	3.240	0.001	0.157	0.638
room_type_reserved_Room_Type 4	0.2901	0.050	5.826	0.000	0.192	0.388
room_type_reserved_Room_Type 5	0.6758	0.191	3.540	0.000	0.302	1.050
room_type_reserved_Room_Type 6	0.9691	0.140	6.942	0.000	0.695	1.243
room_type_reserved_Room_Type 7	1.4181	0.283	5.013	0.000	0.864	1.973
market_segment_type_Corporate	0.8382	0.097	8.678	0.000	0.649	1.027
market_segment_type_Offline	1.7778	0.049	36.633	0.000	1.683	1.873

Handle Multicollinearity (VIF)

	Feature	VIF
0	const	39243123.05620
1	no_of_adults	1.32528
2	no_of_children	1.97236
3	no_of_weekend_nights	1.05803
4	no_of_week_nights	1.08778
5	required_car_parking_space	1.03890
6	lead_time	1.37501
7	arrival_year	1.41994
8	arrival_month	1.26323
9	repeated_guest	1.73257
10	no_of_previous_cancellations	1.35945
11	no_of_previous_bookings_not_canceled	1.61208
12	avg_price_per_room	1.72434
13	no_of_special_requests	1.24271
14	type_of_meal_plan_Meal Plan 2	1.25147
15	type_of_meal_plan_Not Selected	1.25459
16	room_type_reserved_Room_Type 2	1.09386
17	room_type_reserved_Room_Type 3	1.00073
18	room_type_reserved_Room_Type 4	1.34279
19	room_type_reserved_Room_Type 5	1.02220
20	room_type_reserved_Room_Type 6	1.93285
21	room_type_reserved_Room_Type 7	1.05511
22	market_segment_type_Corporate	1.46806

- The Variance Inflation Factor (VIF) was calculated to assess multicollinearity among the predictor variables. All variables showed VIF values well below 5, indicating no significant multicollinearity in the dataset.

Determine optimal threshold using ROC curve



- The ROC curve indicates strong model performance, with an AUC score of 0.87. The curve lies significantly above the diagonal reference line, demonstrating that the model has a high ability to distinguish between canceled and non-canceled bookings. This suggests that the model is effective in predicting booking cancellations

- Although an alternative threshold was explored, the default threshold of 0.5 was retained as it provided better recall, which is more important for identifying booking cancellations

Model Evaluation

```
Accuracy: 0.8012405237767057
Precision: 0.8622943905525094
Recall: 0.8382533825338253
F1 Score: 0.8501039501039501
```

- The Logistic Regression model achieved an accuracy of 80%, with a precision of 86%and a recall of 83%.
- The F1-score of 85% reflects a good balance between precision and recall, making the model suitable for business use

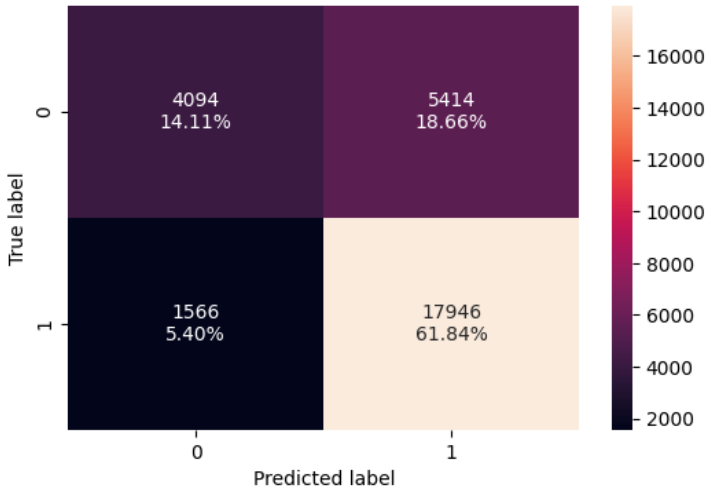
Key insights

- The tuned Logistic Regression model successfully converged and showed improved performance. Most of the variables were found to be statistically significant after removing high p-value features. The model achieved a Pseudo R^2 of 0.3283, indicating moderate explanatory power, and was statistically significant overall.
- Key factors influencing cancellations include lead time, average price per room, and number of nights, which increase the likelihood of cancellation. On the other hand, repeated guests, special requests, and offline bookings significantly reduce the chances of cancellation. These insights can help the hotel implement targeted strategies to minimize cancellation

Decision Tree (Pre-pruning)

Using GridSearch for Hyperparameter tuning of our tree model

Confusion matrix using scikit-learn for the model estimator on X_train and y_train.



Training performance of the classifier estimator on X_train and y_train

	Accuracy	Recall	Precision	F1
0	0.75948	0.91974	0.76824	0.83719

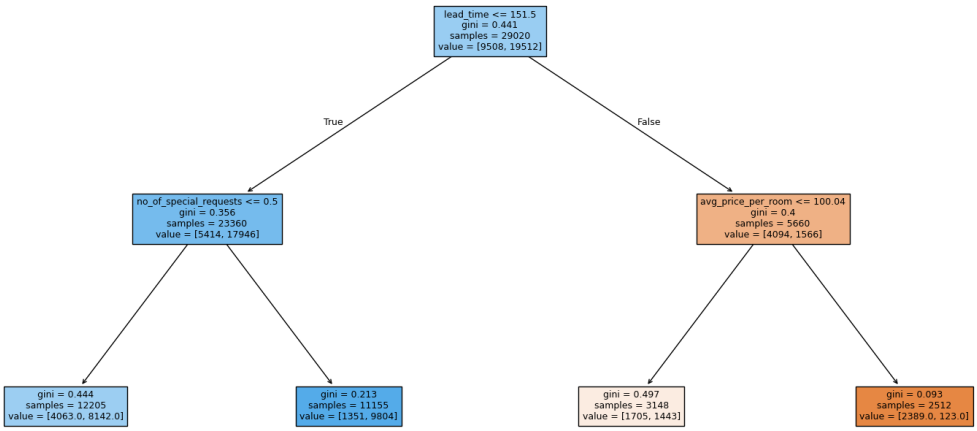
Confusion matrix for the model estimator on X_test and y_test

	Accuracy	Recall	Precision	F1
0	0.76361	0.91533	0.77423	0.83889

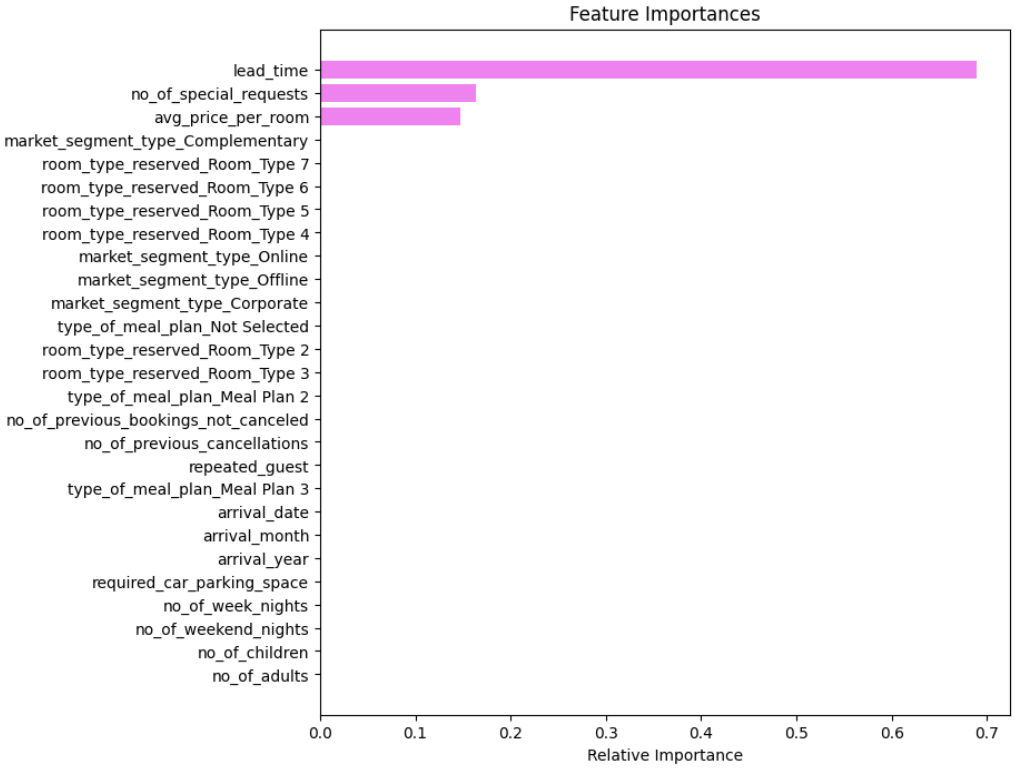
The pre-pruned Decision Tree model achieved an accuracy of 76.5% and a recall of 91.66%, model is very good at identifying actual cases

Feature importances

Decision Tree

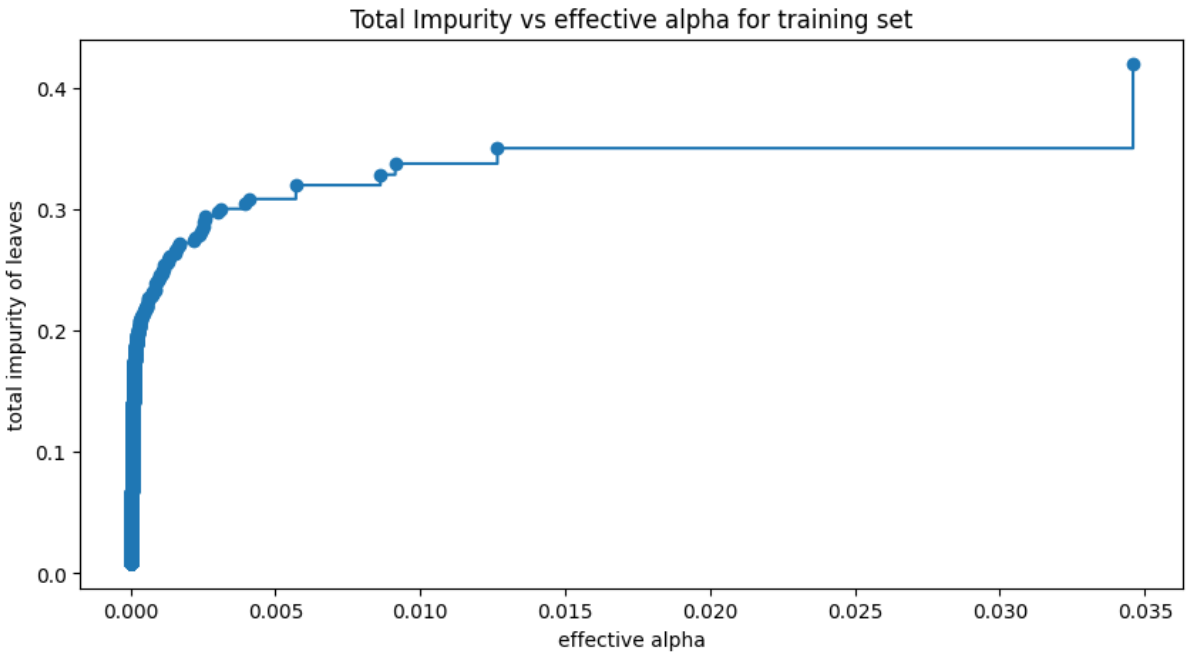


- The Decision Tree model identifies lead_time ≤ 151.5 as the most important factor influencing booking cancellations.
- Bookings made well in advance are more likely to be canceled. Additionally, customers with no_of_special_requests ≤ 0.5 show a higher tendency to cancel, while those with more engagement are less likely to cancel.
- Room price also plays a role, indicating price sensitivity among customers. Overall, the model helps identify high-risk and low-risk customer segments, enabling targeted strategies to reduce cancellations



In the pre tuned decision tree also, lead time, avg price per room,no of special request are the most important features

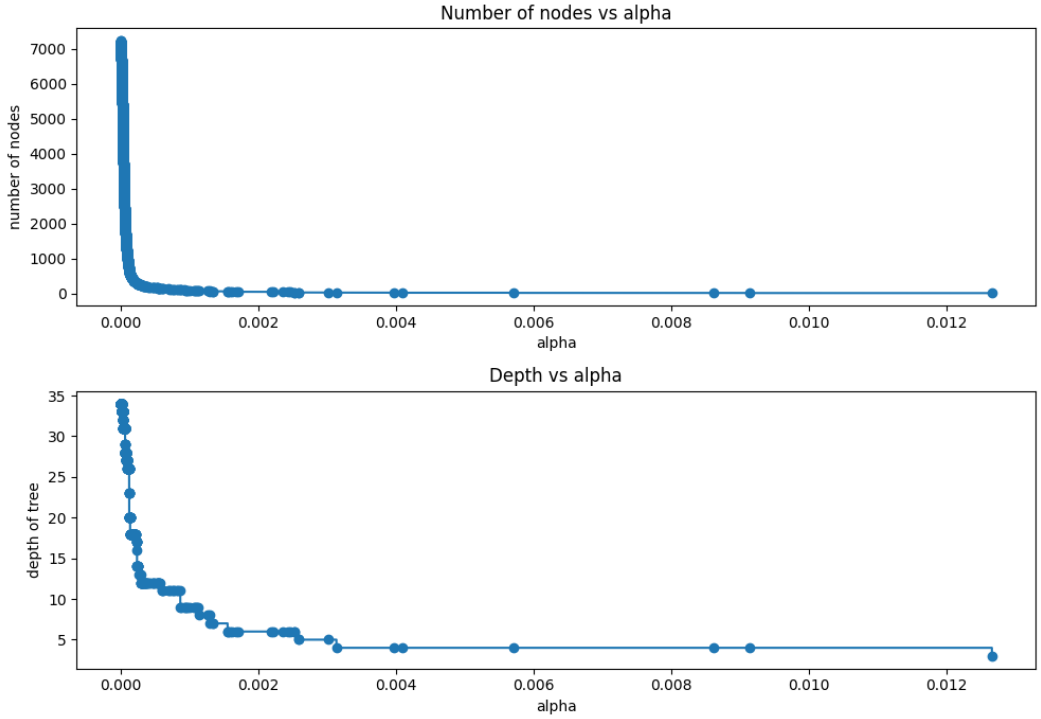
Decision Tree (Post-pruning)



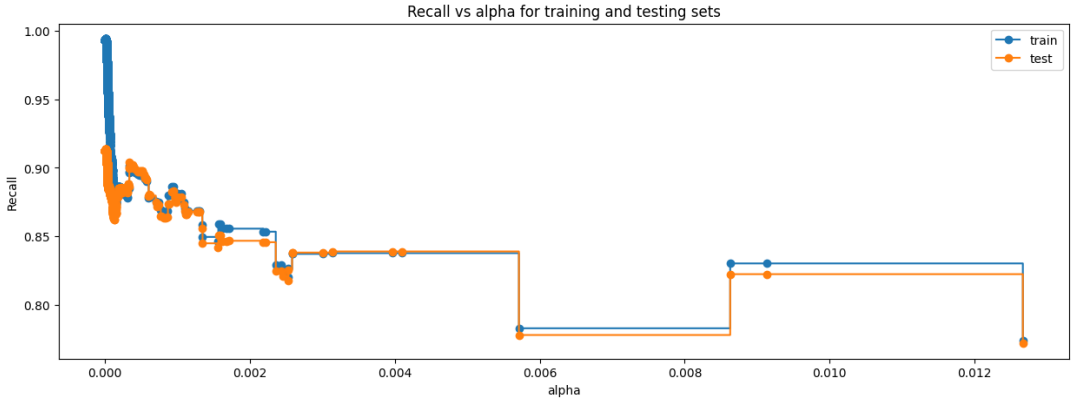
- At very low alpha values (0–0.001), impurity is low (0.01–0.20) indicating overfitting.
- As alpha increases to 0.002–0.01, impurity rises to 0.25–0.35 showing a balanced model. Beyond 0.01, impurity increases to 0.35–0.42 indicating underfitting. Hence, the optimal alpha lies between 0.002 and 0.01

```
Number of nodes in the last tree is: 1 with ccp_alpha: 0.08066551870898075
```

At high alpha value (0.08), the tree reduces to a single node, indicating complete pruning and underfitting

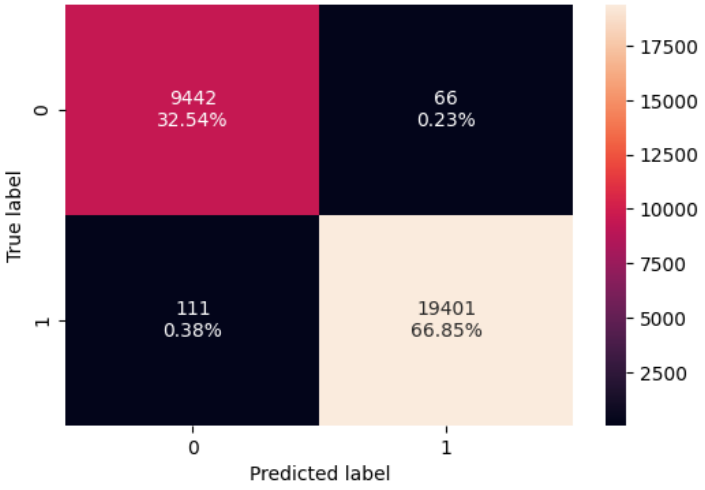


As the cost complexity parameter (alpha) increases, both the number of nodes and the depth of the decision tree decrease significantly. This confirms that alpha effectively controls tree complexity — a small alpha produces a large, overfitted tree, while a larger alpha produces a simpler, more generalizable model.



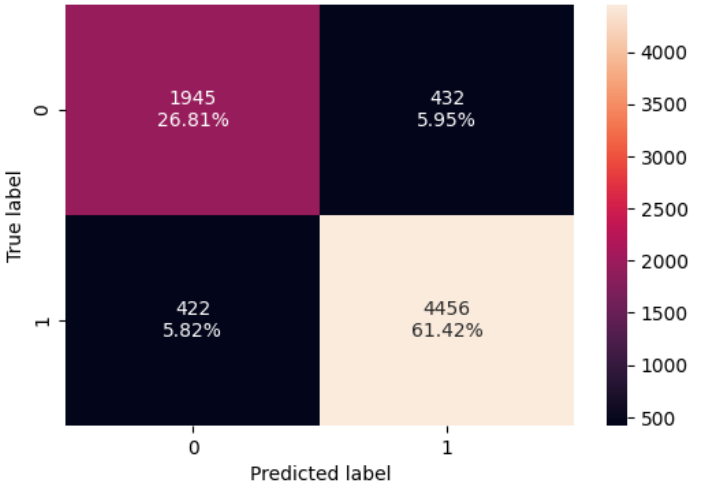
The Recall vs Alpha graph shows that at very low alpha values, the training recall is significantly higher than test recall, indicating overfitting. As alpha increases, the gap closes and both stabilize around 0.83–0.84, suggesting a well-generalized model. Beyond alpha = 0.006, recall begins to drop, indicating underfitting. Therefore, the optimal alpha lies in the range of 0.002 to 0.006

confusion_matrix_sklearn(best_model, X_train, y_train)



	Accuracy	Recall	Precision	F1
0	0.99390	0.99431	0.99661	0.99546

confusion_matrix_sklearn(best_model, X_test, y_test)



	Accuracy	Recall	Precision	F1
0	0.88229	0.91349	0.91162	0.91255

Training performance comparison:

	Decision Tree without class_weight	Decision Tree with class_weight	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	0.99418	0.99363	0.75948	0.99390
Recall	0.99518	0.99344	0.91974	0.99431
Precision	0.99615	0.99707	0.76824	0.99661
F1	0.99567	0.99525	0.83719	0.99546

Test set performance comparison:

	Decision Tree without class_weight	Decision Tree with class_weight	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)
Accuracy	0.87305	0.88146	0.76361	0.88229
Recall	0.89955	0.91226	0.91533	0.91349
Precision	0.91056	0.91151	0.77423	0.91162
F1	0.90502	0.91189	0.83889	0.91255

- Logistic Regression showed stable performance but lower overall accuracy compared to tree models.
- The Decision Tree without class weights achieved very high training performance (F1 ≈ 0.996) but showed overfitting with a drop in test F1 (≈ 0.905). The model with class weights improved test performance (F1 ≈ 0.912) and recall (≈ 0.912).
- The pre-pruned tree reduced overfitting and achieved high recall (≈ 0.915) but had lower overall performance (F1 ≈ 0.839), indicating underfitting.
- The post-pruned Decision Tree achieved the best balance with highest test F1-score (≈ 0.913), along with strong recall (≈ 0.913) and precision (≈ 0.912).
- Hence, the post-pruned Decision Tree is selected as the final model due to its superior performance and better generalization

Actionable insights and recommendations

- Customers with high lead time and no special requests are more likely to cancel, so sending reminders and offering limited-time discounts can improve commitment. Encouraging customers to add special requests can reduce cancellations by increasing engagement.
- Repeat guests show lower cancellation rates, so loyalty programs and exclusive offers should be introduced to retain them. Higher room prices are associated with increased cancellations, suggesting the need for dynamic pricing and flexible offers. Different market segments behave differently, so targeted strategies should be applied, especially for higher-risk segments like online bookings.
- The final model can be used to identify high-risk bookings in advance, enabling proactive actions such as personalized offers and reminders to reduce cancellations and improve overall revenue stability